

Normal Distribution:

A continuous RV X is said to have a normal distribution with mean μ and variance σ^2 ($N(\mu, \sigma^2)$) if it has pdf given by

$$f_X(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2},$$

$$-\infty < x < \infty,$$

$$\mu \in \mathbb{R}, \sigma > 0$$

$$\int_{-\infty}^{\infty} f_X(x) dx = \int_{-\infty}^{\infty} \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} dx$$

$$= \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-\frac{t^2}{2}} dt$$

$$\begin{aligned} \text{Put } \frac{x-\mu}{\sigma} &= t \\ \Rightarrow \frac{1}{\sigma} dx &= dt \end{aligned}$$

$$= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-\frac{t^2}{2}} dt$$

$$= \frac{1}{\sqrt{2\pi}} 2 \int_0^{\infty} e^{-\frac{t^2}{2}} dt$$

$$\begin{aligned} \frac{t^2}{2} &= z \\ \Rightarrow t dt &= dz \end{aligned}$$

$$= \frac{2}{\sqrt{2\pi}} \int_0^{\infty} e^{-z} \frac{1}{\sqrt{2z}} dz$$

$$= \frac{1}{\sqrt{\pi}} \int_0^{\infty} e^{-z} z^{-\frac{1}{2}} dz$$

$$= \frac{1}{\sqrt{\pi}} \int_0^{\infty} e^{-z} z^{\frac{1}{2}-1} dz$$

$$= \frac{1}{\sqrt{\pi}} \Gamma\left(\frac{1}{2}\right) = 1$$

$$X \sim N(\mu, \sigma^2)$$

$$\underline{M_X(t)} = E(e^{tX})$$

$$= \int_{-\infty}^{\infty} e^{tx} \cdot f_X(x) dx$$

$$= \int_{-\infty}^{\infty} e^{tx} \cdot \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} dx$$

$$dz = \frac{1}{\sigma} dx$$

$$z = \frac{x-\mu}{\sigma}$$

$$= \int_{-\infty}^{\infty} e^{t(\mu+\sigma z)} \cdot \frac{1}{\sqrt{2\pi}} e^{-\frac{z^2}{2}} dz$$

$$= e^{\mu t} \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} e^{t\sigma z - \frac{z^2}{2}} dz$$

$$= e^{\mu t} \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}(z^2 - 2t\sigma z)} dz$$

$+ t^2\sigma^2 - t^2\sigma^2$

$$= e^{\mu t} \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}\{(z - t\sigma)^2 - t^2\sigma^2\}} dz$$

$$= e^{\mu t} \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}(z - t\sigma)^2} \cdot e^{\frac{1}{2}\sigma^2 t^2} dz$$

$$= e^{\mu t + \frac{1}{2}\sigma^2 t^2} \cdot \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}(z - t\sigma)^2} dz$$

$$M_X(t) = e^{\mu t + \frac{1}{2}\sigma^2 t^2}$$

✓

$$M'_X(t) = e^{\mu t + \frac{1}{2}\sigma^2 t^2} \cdot (\mu + \sigma^2 t)$$

$$M''_X(t) = e^{\mu t + \frac{1}{2}\sigma^2 t^2} \cdot (\mu + \sigma^2 t)^2 + e^{\mu t + \frac{1}{2}\sigma^2 t^2} \sigma^2$$

□ Let $X \sim N(\mu, \sigma^2)$, $Y = aX + b$, $a \neq 0$, $b \in \mathbb{R}$

$$M_Y(t) = E(e^{tY}) \quad \rightarrow \quad M_X(t) = e^{\mu t + \frac{1}{2}\sigma^2 t^2}$$

$$= E(e^{t(ax+b)})$$

$$= E(e^{tax} \cdot e^{tb})$$

$$= e^{tb} \cdot \underline{E(e^{atx})}$$

$$= e^{bt} M_X(at)$$

$$= e^{bt} \cdot e^{\mu \cdot at + \frac{1}{2}\sigma^2 a^2 t^2}$$

or, $M_Y(t) = e^{(a\mu+b)t + \frac{1}{2}a^2\sigma^2 t^2}$ which is a MGF of

$N(a\mu+b, a^2\sigma^2)$ distribution.

✓ Theorem: (Linearity property of a normal distribution)

If $X \sim N(\mu, \sigma^2)$ and $Y = aX + b$, $a \neq 0$, then $Y \sim N(a\mu+b, a^2\sigma^2)$

Remark: $X \sim N(\mu, \sigma^2)$. $Z = \frac{X-\mu}{\sigma} = \frac{1}{\sigma}X - \frac{\mu}{\sigma}$

$$\sim N\left(\frac{\mu}{\sigma} - \frac{\mu}{\sigma}, \frac{1}{\sigma^2} \cdot \sigma^2\right)$$

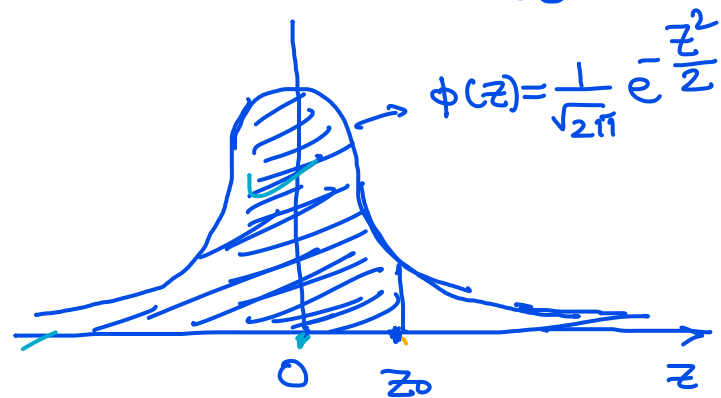
$$\equiv N(0, 1)$$

$N(0,1)$: This is called a standard normal R.V.

The pdf of Z is denoted by $\phi(z) = \frac{1}{\sqrt{2\pi}} e^{-\frac{z^2}{2}}$, $-\infty < z < \infty$

The cdf of Z is denoted by $\Phi(z) = P(Z \leq z) = \int_{-\infty}^z \phi(t) dt$

$$\Phi(z_0) = \int_{-\infty}^{z_0} \frac{1}{\sqrt{2\pi}} e^{-\frac{t^2}{2}} dt$$

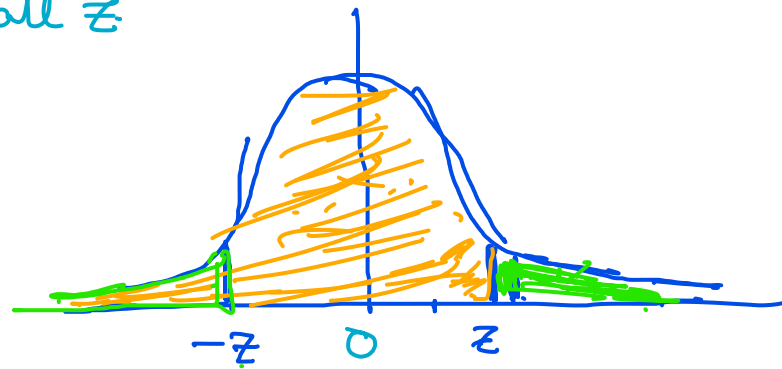


$$\boxed{\Phi(z) + \Phi(-z) = 1} \text{ for all } z$$

$$\Phi(0) + \Phi(0) = 1$$

$$\Rightarrow 2\Phi(0) = 1$$

$$\Rightarrow \Phi(0) = \frac{1}{2}$$



□ Consider $X \sim N(\mu, \sigma^2) \Rightarrow Z = \frac{X - \mu}{\sigma} \sim N(0, 1)$

$$\underline{P(a < X < b)} = P(a - \mu < X - \mu < b - \mu)$$

$$= P\left(\frac{a - \mu}{\sigma} < \frac{X - \mu}{\sigma} < \frac{b - \mu}{\sigma}\right)$$

$$= P\left(\frac{a - \mu}{\sigma} < Z < \frac{b - \mu}{\sigma}\right)$$

$$= \underline{P(a^* < Z < b^*)}$$

$$= P(Z \leq b^*) - P(Z \leq a^*)$$

$$= \Phi(b^*) - \Phi(a^*)$$

$\phi(z)$